

CSCI A411

Operating Systems

Homework Assignment #1

1. Create a diagram that illustrates the four main components of a computer system. In your diagram, label each component and briefly describe its role. Submit your diagram along with a short paragraph explaining the importance of each component.
2. Investigate a specific multi-core processor. Summarize its architecture, how it handles multitasking, and why this is beneficial in modern computing.
3. Using a virtualization tool like VirtualBox, set up a virtual machine running a different operating system than your main one (e.g., Linux on a Windows machine). Document the steps you took (with two screen shots), the challenges you faced, and what you learned from the experience.